

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name **Berusynth CU 250**
Unique formula identifier (UFI) **WG51-903R-W00J-Y696**

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses Industrial uses: uses of substances as such or in preparations at industrial sites
Lubricant

1.3 Details of the supplier of the safety data sheet

Carl Bechem GmbH
Weststr. 120
58089 Hagen
Germany

Telephone: +49 2331 935 0
e-mail: ps@bechem.com
Website: <http://www.bechem.com>

Additional information

Manufacturer					
Country	Name	Postal code	City	Telephone	Website
Germany	CARL BECHEM GMBH	58089	Hagen	49 2331 935 0	

e-Mail address of competent person responsible for the SDS ps@bechem.com (Produktsicherheit)

1.4 Emergency telephone number

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP)

Section	Hazard class	Category	Hazard class and category	Hazard statement
4.1C	hazardous to the aquatic environment - chronic hazard	3	Aquatic Chronic 3	H412

For full text of abbreviations: see SECTION 16.

The most important adverse physicochemical, human health and environmental effects
Spillage and fire water can cause pollution of watercourses.

2.2 Label elements

Labelling
- Signal word not required

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

- Pictograms not required
- Hazard statements
H412 Harmful to aquatic life with long lasting effects.
- Precautionary statements
P273 Avoid release to the environment.
P501 Dispose of contents/container to industrial combustion plant.

2.3 Other hazards

2.3.2 Results of PBT and vPvB assessment

Does not contain a PBT-/vPvB-substance at a concentration of $\geq 0,1\%$.

Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) at a concentration of $\geq 0,1\%$.

SECTION 3: Composition/information on ingredients

3.1 Substances

Not relevant (mixture)

3.2 Mixtures

Description of the mixture

Name of substance	Identifier	Wt%	Classification acc. to GHS	Notes	M-Factors
O,O,O-tris(2(or 4)-C9-10-isoalkylphenyl) phosphorothioate	CAS No 126019-82-7 EC No 406-940-1 Index No 015-171-00-7 REACH Reg. No 01-0000015643-71- xxxx 01-2119930067-42- xxxx	2,5 – < 5	Repr. 2 / H361d Aquatic Chronic 2 / H411		
reaction mass of iso- mers of: C7-9-alkyl 3- (3,5-di-tert-butyl-4-hy- droxyphenyl)propion- ate	CAS No 125643-61-0 EC No 406-040-9 Index No 607-530-00-7 REACH Reg. No 01-0000015551-76- xxxx 01-2119878226-29- xxxx	1 – < 2,5	Aquatic Chronic 4 / H413	GHS-HC	
Oleoylsarcosine	CAS No 110-25-8 REACH Reg. No 01-2119488991-20- xxxx	0,25 – < 1	Acute Tox. 4 / H332 Skin Irrit. 2 / H315 Eye Dam. 1 / H318 Aquatic Acute 1 / H400 Aquatic Chronic 3 / H412		

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Notes

GHS-HC: Harmonised classification (the classification of the substance corresponds to the entry in the list according to 1272/2008/EC, Annex VI)

Name of substance	Specific Conc. Limits	M-Factors	ATE	Exposure route
Oleoylsarcosine	-	-	11 mg _i /4h 1,37 mg _i /4h	inhalation: vapour inhalation: dust/mist

Remarks

For full text of abbreviations: see SECTION 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

General notes

Do not leave affected person unattended. Remove victim out of the danger area. Keep affected person warm, still and covered. Take off immediately all contaminated clothing. In all cases of doubt, or when symptoms persist, seek medical advice. In case of unconsciousness place person in the recovery position. Never give anything by mouth.

Following inhalation

If breathing is irregular or stopped, immediately seek medical assistance and start first aid actions. Provide fresh air.

Following skin contact

Rinse skin with water/shower. Take off contaminated clothing.

Following eye contact

Remove contact lenses, if present and easy to do. Continue rinsing. Irrigate copiously with clean, fresh water for at least 10 minutes, holding the eyelids apart.

Following ingestion

Rinse mouth with water (only if the person is conscious). Do NOT induce vomiting.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms and effects are not known to date.

4.3 Indication of any immediate medical attention and special treatment needed

none

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media

Water spray, BC-powder, Carbon dioxide (CO₂)

Unsuitable extinguishing media

Water jet

5.2 Special hazards arising from the substance or mixture

Hazardous combustion products

Carbon monoxide (CO), Carbon dioxide (CO₂), Sulphur dioxide (SO₂)

5.3 Advice for firefighters

In case of fire and/or explosion do not breathe fumes. Co-ordinate firefighting measures to the fire surroundings. Do not allow firefighting water to enter drains or water courses. Collect contaminated firefighting water separately. Fight fire with normal precautions from a reasonable distance.

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel

Remove persons to safety.

For emergency responders

Wear breathing apparatus if exposed to vapours/dust/spray/gases.

6.2 Environmental precautions

Keep away from drains, surface and ground water. Retain contaminated washing water and dispose of it. If substance has entered a water course or sewer, inform the responsible authority.

6.3 Methods and material for containment and cleaning up

Advice on how to contain a spill

Covering of drains

Advice on how to clean up a spill

Wipe up with absorbent material (e.g. cloth, fleece). Collect spillage: sawdust, kieselgur (diatomite), sand, universal binder

Appropriate containment techniques

Use of adsorbent materials.

Other information relating to spills and releases

Place in appropriate containers for disposal. Ventilate affected area.

6.4 Reference to other sections

Hazardous combustion products: see section 5. Personal protective equipment: see section 8. Incompatible materials: see section 10. Disposal considerations: see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Personal protective equipment: see section 8.

Recommendations

- Measures to prevent fire as well as aerosol and dust generation

Use local and general ventilation. Use only in well-ventilated areas.

Advice on general occupational hygiene

Wash hands after use. Do not eat, drink and smoke in work areas. Remove contaminated clothing and protective equipment before entering eating areas. Never keep food or drink in the vicinity of chemicals. Never place chemicals in containers that are normally used for food or drink. Keep away from food, drink and animal feedingstuffs.

7.2 Conditions for safe storage, including any incompatibilities

There is no additional information.

Concerning incompatibility: see below "Conditions to avoid" and "Incompatible materials" (Section 10).

7.3 Specific end use(s)

See section 16 for a general overview.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

This information is not available.

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Relevant DNELs of components						
Name of substance	CAS No	Endpoint	Threshold level	Protection goal, route of exposure	Used in	Exposure time
O,O,O-tris(2(or 4)-C9-10-isoalkylphenyl) phosphorothioate	126019-82-7	DNEL	11,75 mg/m ³	human, inhalatory	worker (industry)	chronic - systemic effects
O,O,O-tris(2(or 4)-C9-10-isoalkylphenyl) phosphorothioate	126019-82-7	DNEL	33,3 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	DNEL	3 mg/m ³	human, inhalatory	worker (industry)	chronic - systemic effects
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	DNEL	8,6 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects
Oleoylsarcosine	110-25-8	DNEL	0,8 mg/m ³	human, inhalatory	worker (industry)	chronic - systemic effects
Oleoylsarcosine	110-25-8	DNEL	20 mg/kg bw/day	human, dermal	worker (industry)	chronic - systemic effects

8.2 Exposure controls

Appropriate engineering controls

General ventilation.

Individual protection measures (personal protective equipment)

Eye/face protection

Wear eye/face protection.

Skin protection

- Hand protection

Wear suitable gloves. Chemical protection gloves are suitable, which are tested according to EN 374. Check leak-tightness/impermeability prior to use. For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

- Type of material

IIR: isobutene-isoprene (butyl) rubber

- Material thickness

> 1 mm.

- Breakthrough times of the glove material

>10 minutes (permeation: level 1)

- Not suitable are gloves made of the following materials

Fabric, Leather

- Other protection measures

Take recovery periods for skin regeneration. Preventive skin protection (barrier creams/ointments) is recommended. Wash hands thoroughly after handling.

Respiratory protection

In case of inadequate ventilation wear respiratory protection.

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Environmental exposure controls

Use appropriate container to avoid environmental contamination. Keep away from drains, surface and ground water.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	liquid
Colour	light yellow
Odour	characteristic
Melting point/freezing point	not determined
Boiling point or initial boiling point and boiling range	not determined
Flammability	this material is combustible, but will not ignite readily
Lower and upper explosion limit	not determined
Flash point	≥250 °C (o.c.)
Auto-ignition temperature	not determined
pH (value)	not applicable
Kinematic viscosity	225 – 275 mm ² /s at 40 °C
Water solubility	insoluble

Partition coefficient

Partition coefficient n-octanol/water (log value)	this information is not available
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Vapour pressure	not determined
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Density and/or relative density

Density	0,936 – 0,945 g/cm ³ at 15 °C
Relative vapour density	information on this property is not available

Particle characteristics	not relevant (liquid)
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9.2 Other information

Information with regard to physical hazard	hazard classes acc. to GHS (physical hazards):
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Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

classes	not relevant
Other safety characteristics	there is no additional information

SECTION 10: Stability and reactivity

10.1 Reactivity

Concerning incompatibility: see below "Conditions to avoid" and "Incompatible materials".

10.2 Chemical stability

See below "Conditions to avoid".

10.3 Possibility of hazardous reactions

No known hazardous reactions.

10.4 Conditions to avoid

There are no specific conditions known which have to be avoided.

10.5 Incompatible materials

Oxidisers

10.6 Hazardous decomposition products

Reasonably anticipated hazardous decomposition products produced as a result of use, storage, spill and heating are not known. Hazardous combustion products: see section 5.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Test data are not available for the complete mixture.

Classification procedure

The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

Classification acc. to GHS

Acute toxicity

Shall not be classified as acutely toxic.

Acute toxicity estimate (ATE) of components			
Name of substance	CAS No	Exposure route	ATE
Oleoylsarcosine	110-25-8	inhalation: vapour	11 mg/l/4h
Oleoylsarcosine	110-25-8	inhalation: dust/mist	1,37 mg/l/4h

Skin corrosion/irritation

Shall not be classified as corrosive/irritant to skin.

Serious eye damage/eye irritation

Shall not be classified as seriously damaging to the eye or eye irritant.

Respiratory or skin sensitisation

Shall not be classified as a respiratory or skin sensitiser.

Germ cell mutagenicity

Shall not be classified as germ cell mutagenic.

Carcinogenicity

Shall not be classified as carcinogenic.

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Reproductive toxicity

Shall not be classified as a reproductive toxicant.

Specific target organ toxicity - single exposure

Shall not be classified as a specific target organ toxicant (single exposure).

Specific target organ toxicity - repeated exposure

Shall not be classified as a specific target organ toxicant (repeated exposure).

Aspiration hazard

Shall not be classified as presenting an aspiration hazard.

11.2 Information on other hazards

Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) at a concentration of $\geq 0,1\%$.

Other information

There is no additional information.

SECTION 12: Ecological information

12.1 Toxicity

Harmful to aquatic life with long lasting effects.

Aquatic toxicity (chronic) of components					
Name of substance	CAS No	Endpoint	Value	Species	Exposure time
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	LC50	$>100 \text{ mg/l}$	fish	14 d
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	EC50	$>100 \text{ mg/l}$	microorganisms	3 h
Oleoylsarcosine	110-25-8	EC50	1.300 mg/l	microorganisms	3 h

12.2 Persistence and degradability

Degradability of components						
Name of substance	CAS No	Process	Degradation rate	Time	Method	Source
O,O,O-tris(2(or 4)-C9-10-isoalkyl-phenyl) phosphorothioate	126019-82-7	carbon dioxide generation	2 %	28 d		ECHA
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	carbon dioxide generation	4 %	28 d		ECHA
Oleoylsarcosine	110-25-8	carbon dioxide generation	85,2 %	28 d		ECHA

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

12.3 Bioaccumulative potential

Data are not available.

Name of substance	CAS No	BCF	Log KOW	BOD5/COD
O,O,O-tris(2(or 4)-C9-10-isoalkyl-phenyl) phosphorothioate	126019-82-7		20 (20 °C)	
reaction mass of isomers of: C7-9-alkyl 3-(3,5-di-tert-butyl-4-hydroxyphenyl)propionate	125643-61-0	258	9,2 (25 °C)	

12.4 Mobility in soil

Data are not available.

12.5 Results of PBT and vPvB assessment

Does not contain a PBT-/vPvB-substance at a concentration of $\geq 0,1\%$.

12.6 Endocrine disrupting properties

Does not contain an endocrine disruptor (ED) at a concentration of $\geq 0,1\%$.

12.7 Other adverse effects

Data are not available.

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Sewage disposal-relevant information

Do not empty into drains. Avoid release to the environment. Refer to special instructions/safety data sheets.

Waste treatment of containers/packages

Completely emptied packages can be recycled. Handle contaminated packages in the same way as the substance itself.

Remarks

Please consider the relevant national or regional provisions. Waste shall be separated into the categories that can be handled separately by the local or national waste management facilities.

SECTION 14: Transport information

- | | |
|--|---|
| 14.1 UN number or ID number | not subject to transport regulations |
| 14.2 UN proper shipping name | not relevant |
| 14.3 Transport hazard class(es) | none |
| 14.4 Packing group | not assigned |
| 14.5 Environmental hazards | non-environmentally hazardous acc. to the dangerous goods regulations |
| 14.6 Special precautions for user | There is no additional information. |
| 14.7 Maritime transport in bulk according to IMO instruments | The cargo is not intended to be carried in bulk. |

Information for each of the UN Model Regulations

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Transport information - National regulations - Additional information (UN RTDG)

Not subject to transport regulations: UN RTDG

International Maritime Dangerous Goods Code (IMDG) - Additional information

Not subject to IMDG.

International Civil Aviation Organization (ICAO-IATA/DGR) - Additional information

Not subject to ICAO-IATA.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

There is no additional information.

Relevant provisions of the European Union (EU)

Restrictions according to REACH, Annex XVII

Dangerous substances with restrictions (REACH, Annex XVII)				
Name of substance	Name acc. to inventory	CAS No	EC No	No
Berusynth CU 250	this product meets the criteria for classification in accordance with Regulation No 1272/2008/EC			3

List of substances subject to authorisation (REACH, Annex XIV) / SVHC - candidate list

none of the ingredients are listed

Seveso Directive

2012/18/EU (Seveso III)			
No	Dangerous substance/hazard categories	Qualifying quantity (tonnes) for the application of lower and upper-tier requirements	Notes
	not assigned		

Directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS)

none of the ingredients are listed

Regulation on persistent organic pollutants (POP)

none of the ingredients are listed

15.2 Chemical safety assessment

Chemical safety assessments for substances in this mixture were not carried out.

SECTION 16: Other information

Indication of changes (revised safety data sheet)

Section	Former entry (text/value)	Actual entry (text/value)	Safety-relevant
2.2		- Precautionary statements: change in the listing (table)	yes

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Abbreviations and acronyms

Abbr.	Descriptions of used abbreviations
Acute Tox.	Acute toxicity
Aquatic Acute	Hazardous to the aquatic environment - acute hazard
Aquatic Chronic	Hazardous to the aquatic environment - chronic hazard
ATE	Acute Toxicity Estimate
BCF	Bioconcentration factor
BOD	Biochemical Oxygen Demand
CAS	Chemical Abstracts Service (service that maintains the most comprehensive list of chemical substances)
CLP	Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures
COD	Chemical oxygen demand
DGR	Dangerous Goods Regulations (see IATA/DGR)
DNEL	Derived No-Effect Level
EC50	Effective Concentration 50 %. The EC50 corresponds to the concentration of a tested substance causing 50 % changes in response (e.g. on growth) during a specified time interval
EC No	The EC Inventory (EINECS, ELINCS and the NLP-list) is the source for the seven-digit EC number, an identifier of substances commercially available within the EU (European Union)
ED	Endocrine disruptor
EINECS	European Inventory of Existing Commercial Chemical Substances
ELINCS	European List of Notified Chemical Substances
Eye Dam.	Seriously damaging to the eye
Eye Irrit.	Irritant to the eye
GHS	"Globally Harmonized System of Classification and Labelling of Chemicals" developed by the United Nations
IATA	International Air Transport Association
IATA/DGR	Dangerous Goods Regulations (DGR) for the air transport (IATA)
ICAO	International Civil Aviation Organization
IMDG	International Maritime Dangerous Goods Code
index No	The Index number is the identification code given to the substance in Part 3 of Annex VI to Regulation (EC) No 1272/2008
LC50	Lethal Concentration 50%: the LC50 corresponds to the concentration of a tested substance causing 50 % lethality during a specified time interval
log KOW	n-Octanol/water
NLP	No-Longer Polymer
PBT	Persistent, Bioaccumulative and Toxic
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
Repr.	Reproductive toxicity
Skin Corr.	Corrosive to skin
Skin Irrit.	Irritant to skin
SVHC	Substance of Very High Concern

Safety Data Sheet

acc. to Regulation (EC) No. 1907/2006 (REACH)



Version number: GHS 3.2
Replaces version of: 21.01.2022 (GHS 2)

Revision: 03.12.2025

Abbr.	Descriptions of used abbreviations
UN RTDG	UN Recommendations on the Transport of Dangerous Good
vPvB	Very Persistent and very Bioaccumulative

Key literature references and sources for data

UN Recommendations on the Transport of Dangerous Good. International Maritime Dangerous Goods Code (IMDG). Dangerous Goods Regulations (DGR) for the air transport (IATA).

Classification procedure

Physical and chemical properties: The classification is based on tested mixture.

Health hazards, Environmental hazards: The method for classification of the mixture is based on ingredients of the mixture (additivity formula).

List of relevant phrases (code and full text as stated in section 2 and 3)

Code	Text
H315	Causes skin irritation.
H318	Causes serious eye damage.
H332	Harmful if inhaled.
H361d	Suspected of damaging the unborn child.
H400	Very toxic to aquatic life.
H411	Toxic to aquatic life with long lasting effects.
H412	Harmful to aquatic life with long lasting effects.
H413	May cause long lasting harmful effects to aquatic life.

Disclaimer

This information is based upon the present state of our knowledge. This SDS has been compiled and is solely intended for this product.